IE301 · TOPIC 1 · FINAL PRESENTATION

Operational analysis of a day-care chemotherapy unit

Simulation, evening optimiser and results

Team INFUCINQ
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Simulation
many linked days of the unit

Evening optimiser
books tomorrow's chairs and doctors

Six experiments
120 days each, same patients

WHERE WE ARE

From the Tata Memorial visit to a working model

300–400

patients a day at TMH

70–80

chairs, slots booked by hand

2–3 days

lost when a slot is missed

What we built

- A simulation of the unit over many linked days: cycles, unfit patients, rebooking
- An evening optimiser that books tomorrow's chairs and doctors
- Six experiments: booking rules, pre-clearance, reserved chairs, nurses, robustness, 24/7; 120 days each
- A website that plays any day minute by minute

Demand
8 new patients a day, 5–8 cycles each

→

Optimiser
arrival times, chair slots, doctors

→

Simulated day
random stage times, queues, unfit

→

Scorecard
chairs, doctors, sent home, limits

Repeated every evening, for 30 warm-up + 120 reported days

The model is a generic Indian public unit (20 chairs). TMH is where we observe; every input is a placeholder until TMH data.

PATIENT FLOW

Who comes, and what they are told

Patients are never told a chair time, only when to arrive

Pre-clearance visit

day −1

Register → Blood → **Lab** 100–200 min → Doctor: fit or unfit → Leave: chemo tomorrow

Same-day patient

far + some local

Register → **Blood + lab** 100–200 min → Doctor: fit or unfit → Billing + drug: 40–75 min → Chair: 30 min – 10 h

Pre-cleared patient

local only

Register → Billing: 10–20 min → Drug: 25–60 min → **Chair** 30 min – 10 h

Urgent rebook

Missed a cycle (unfit, then fit; or sent home). Skips the long chair queue. Uses reserved chairs.

Standby

Called extra to fill chairs left by no-shows. Seated only after every planned patient.

No-chair (10%)

Injection only: nurse setup, no chair. Known after the doctor review.

Pre-clearance lets the unit know, the evening before, who is fit and how long their drip is.

INPUTS

Inputs and distributions

Stage ranges from Presentation 3 · all placeholders until TMH data

Stage	Range	Distribution	Mean
Arrival vs told time	±30 min	Triangular	0 min
Registration	5–15 min	Triangular	10 min
Blood draw	5–10 min	Triangular	7.5 min
Lab result	**100–200 min**	Triangular	**163 min**
Doctor review	15–40 min	Lognormal (median 24.5)	25 min
Billing / insurance	10–20 min	Normal (15 ± 2.5)	15 min
Drug preparation	25–60 min	Lognormal (median 38.7)	40 min
Line + pre-meds	20–40 min	Triangular	30 min
Infusion	30 min – 10 h	Drug mix × lognormal drift	Drug mix

Demand

- 8 new patients a day, 5–8 cycles, 2/3/4 weeks apart
- ≈ 52 cycles reach their day
- 40% far · 10% no-chair · 5% no-show
- 8% unfit · 2% pass away per cycle

Why lognormal

Most reviews and preparations are quick, a few run long. Median = √(low × high); the range covers 9 in 10 patients.

Doctor: $\sqrt{(15 \times 40)} = 24.5$ min
Drug: $\sqrt{(25 \times 60)} = 38.7$ min

Same random numbers for every policy: each rule is tested on exactly the same patients.

OBJECTIVE

What a good day means

Team decision: use chairs and doctors fully; patient waiting is not the goal

Score = 0.60 × chair use + 0.30 × doctor use − 0.10 × share sent home

All three on a 0–1 scale. Chairs and doctors carry 90% of the weight, chairs twice as much as doctors.

Hard limits: a plan that breaks one is out

Overtime
≤ 30 min past the effective close

90th-percentile wait
≤ 6 h, arrival to drip

Cycle rule
every missed cycle given within 3 days

Closing rule



A seated patient always finishes. 18:00 stretches only for far patients; a local patient may use a free chair if they finish before the far patient does. Otherwise: sent home, urgent rebook.

Also: at most 5 patients told the same time; nurses are a variable, the model reports how many are needed.

FORMULATION

The optimiser

Integer programme, solved every evening for tomorrow (HiGHS, 15-min slots)

Decides

- Each patient's chair slot (internal)
- Who is left out of the plan
- Doctor slots for same-day reviews and tomorrow's pre-clearance visits
- Told arrival = slot − buffer

Constraints

- Chairs ≤ 20; reserved chairs kept for urgent
- Nurses: 4 × setups + drips ≤ 4 × N
- Doctors: reviews ≤ what 2 doctors can finish
- Ends by the close; ≤ 5 told the same time
- Never leave out more urgent patients than greedy

Maximises

- 0.60 × chair-minutes used
- + 0.30 × doctor minutes used
- − 0.10 × patients left out
- tiny tie-break: earlier finishes

The buffer Time a patient needs to become ready, set so 8 in 10 are ready by their slot: pre-cleared and urgent 1 h 30 m · same-day 5 h

Checked every constraint re-verified on 20 days (20/20) · equals brute force on 6 small cases (6/6) · never below greedy on its own plan (20/20)

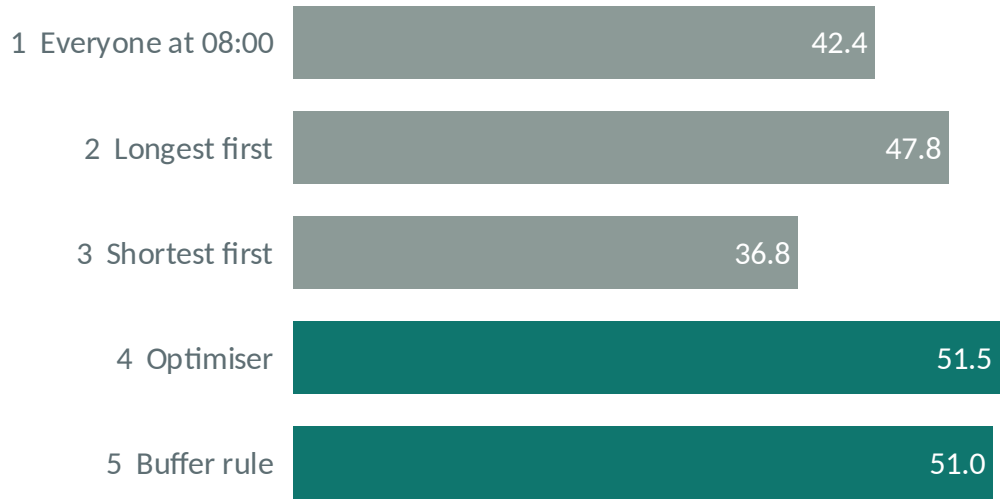
Example: chair slot 11:30, same-day buffer 5 h → patient told to come at 06:30.

RESULT 1 · BOOKING RULES

Does a better rule beat today's practice?

Five ways to tell patients when to come · 120 days each, same patients for every rule

Chair use, % of 08:00–18:00 (higher is better)



Sent home per day, no chair in time (lower is better)



1 Everyone at 08:00 today's practice: all told 08:00, first come first served

2–3 Longest / shortest first longest (or shortest) drips go first; told 1 h before

4 Optimiser same rule, but the evening optimiser picks the best slots

5 Buffer rule told time = chair slot − time needed to get ready

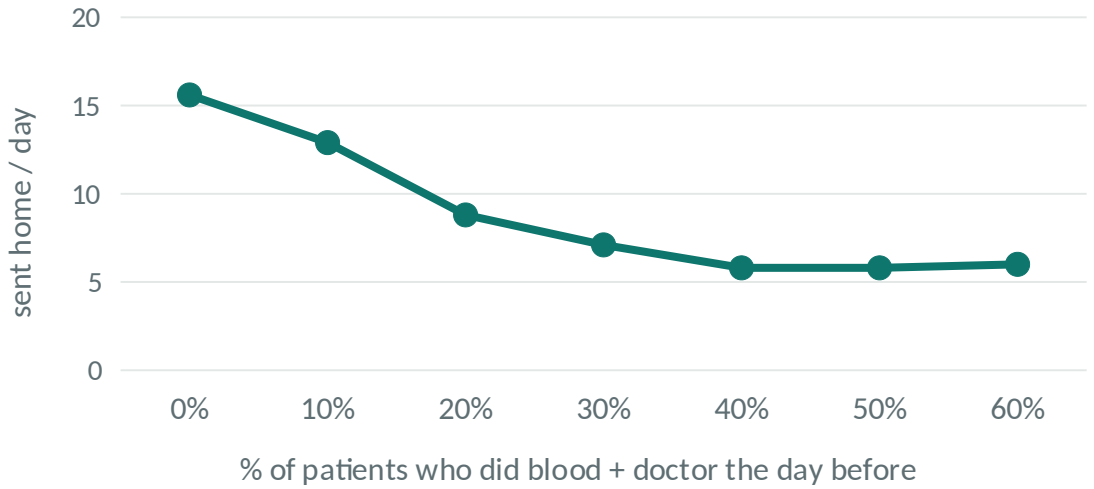
Telling each patient a time based on their own readiness: +9 points chair use and ~22 fewer sent home a day. The simple buffer rule does as well as the optimiser.

RESULT 2 · WHAT THE UNIT CAN CHANGE

Pre-clearance and nurses

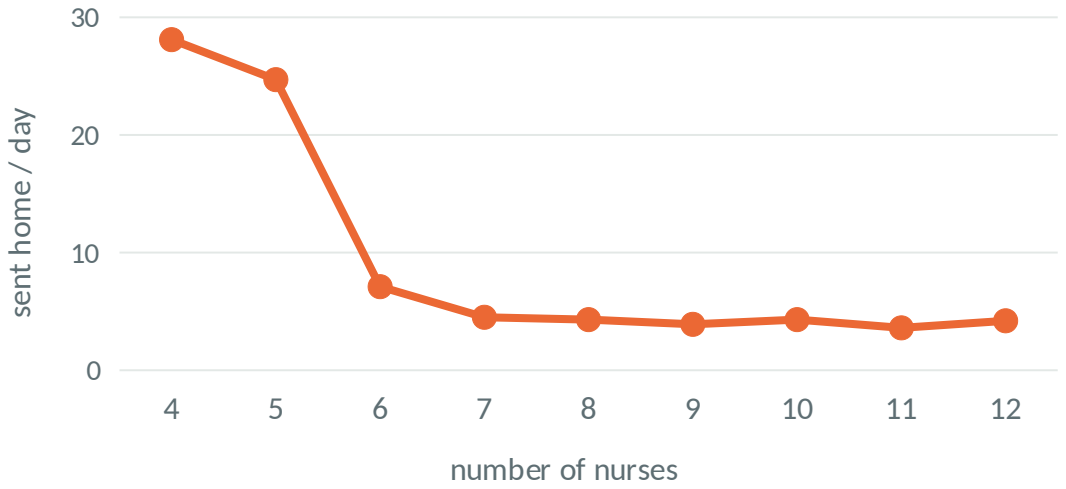
Each point = mean of 120 days, buffer rule + optimiser

More patients pre-cleared → fewer sent home



0% → 60% pre-cleared: sent home 15.6 → 6.0 a day. Chair use barely moves (51%).

More nurses → fewer sent home, up to 7–8



Below 6 nurses the unit breaks; above 8 extra nurses change almost nothing.

Best change the unit controls: pre-clear 60% + 7 nurses → 2.9 sent home a day.

Also tested: reserved chairs and standby patients matter little once the pharmacy has 4 hoods (best: 3 reserved, 5 standby). Robustness: the conclusions hold when every input is halved or doubled.

RESULT 3 · BOTTLENECK

The hidden bottleneck is the pharmacy

Found while checking why the cycle rule failed

3 hoods

12%

of missed cycles given in 3 days

38

sent home a day

4 hoods

99.6%

of missed cycles given in 3 days

7

sent home a day

Why

- ~50 drugs a day × ~40 min = 2,000 hood-minutes; 3 hoods give only 1,800
- Urgent patients waited ~4 h 30 m at the pharmacy, missed the close, were sent home again
- Reserved chairs (best 16%) and extra nurses did not fix it; one more hood did
- 20 missed cycles traced end to end: the rebooking logic is correct

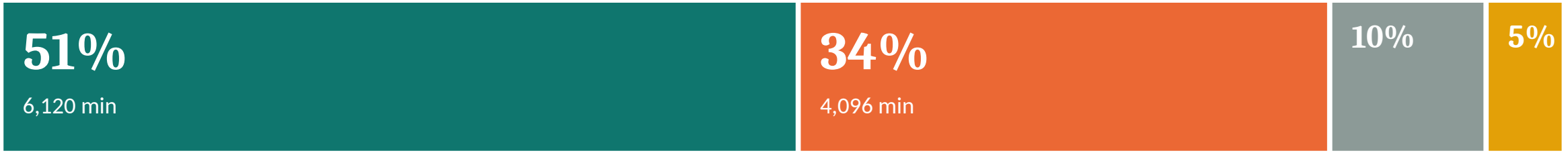
Why chair use stays near 51% Demand, not the schedule: 8 new patients a day fill only 6,120 of 12,000 chair-minutes. Idle time is mostly mornings, while same-day patients are still in lab. For 85% use the unit needs ~13 new patients a day, then a 3rd doctor and a 5th hood.

One extra pharmacy hood matters more than extra chairs, nurses or reserved slots.

RESULT 4 · CHAIR TIME

Where the chair time goes

One day: 20 chairs × 10 h = 12,000 chair-minutes · base case, mean of 120 days



Not ready yet: 34%

Mostly mornings, while same-day patients are still in lab (~3 h), doctor and pharmacy.

Fix: pre-clear patients the day before; faster lab results

Nobody left to seat: 10%

8 new patients a day is not enough to fill 20 chairs.

Fix: more patients (≈13 a day for 85% use), or fewer chairs

No free nurse: 5%

A chair is free and a patient ready, but every nurse is busy setting up.

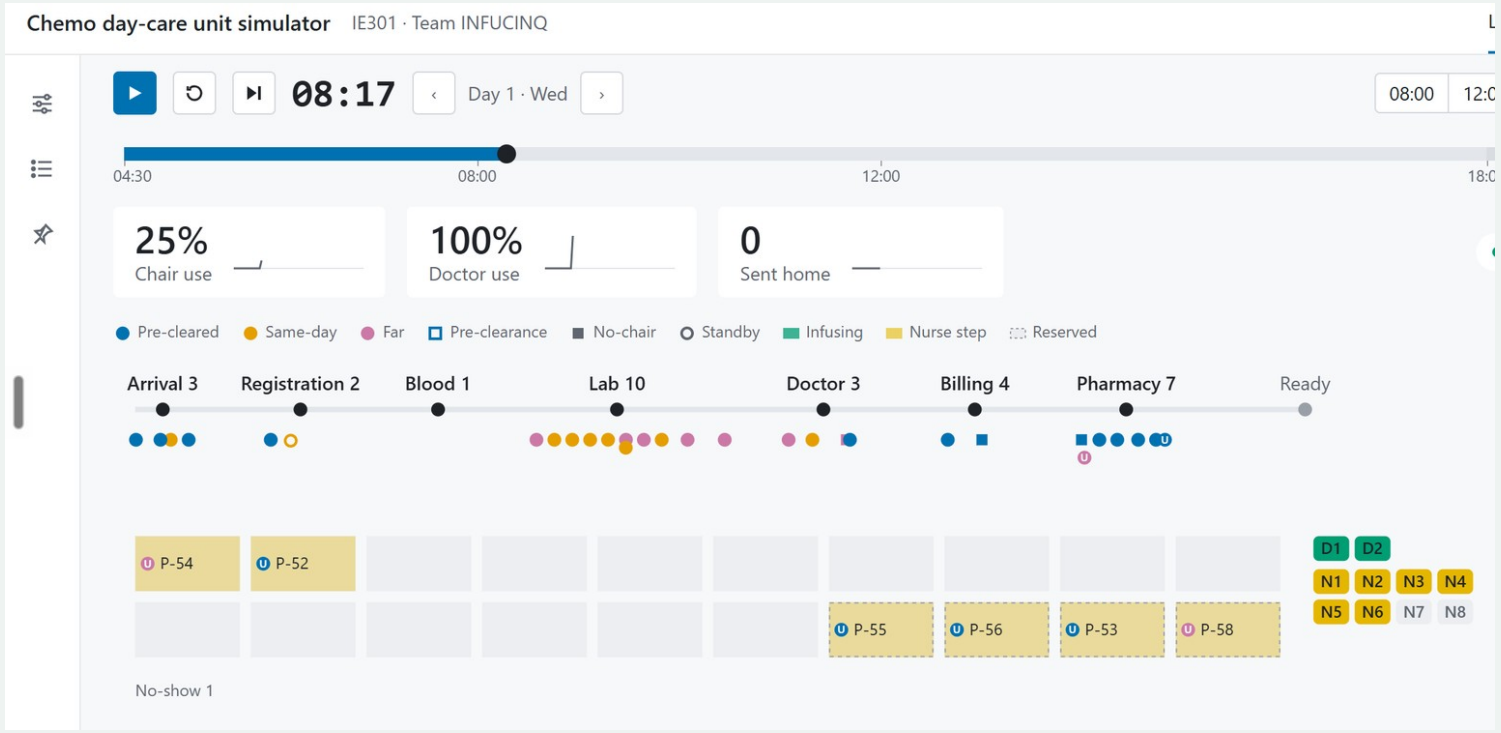
Fix: 7 nurses instead of 6

Chairs mostly wait for patients to become ready. Getting patients ready earlier is the biggest lever.

LIVE DEMO

The simulator

Every day the model plays out, minute by minute, in the browser



Live day

Clock, chair use, doctor use, sent home; station queues; 20 chair tiles that fill as the drip runs; doctor and nurse chips

Experiments

All six experiments, each with one chart and a one-line answer

Model

Flow, closing rule, every input with its source; Python and website give identical numbers

https://ie301-infucinq.iitbresume.workers.dev/

WHAT WE FOUND

Four answers from 120 simulated days

+9 pts

chair use, and ~22 fewer sent home a day, by telling each patient a time based on their own readiness

2.9

sent home a day with 60% pre-cleared and 7 nurses: the best change the unit controls

12% → 99.6%

missed cycles given within 3 days, from one extra pharmacy hood

34%

of chair time is lost to patients who are here but not ready: the biggest lever

TEAM INFUCINQ · IE301 OPERATIONS ANALYSIS · TOPIC 1

Thank you

Questions

Parav · Sahasra · Siddhesh · Vaibhav · Kartikya